Improving Social Connection Through Emotional Intelligence

Bradley Allen

Department of Psychology, Willamette University

PSYC 431W: Reform Psychology Seminar

May 2, 2025

Abstract

For many Americans, a lack of social connection contributes to a sense of loneliness and isolation. Loneliness and isolation are associated with negative health, economic, and educational outcomes. According to the Office of the Surgeon General (2023), more Americans today experience feelings of loneliness and isolation from declining social connections. This paper argues that declining social connections can be reversed by improving emotional intelligence in early childhood education. Schools across the country have successfully increased emotional intelligence through the employment of social emotional learning (SEL) models. Decades of research have revealed SEL models not only improve emotional intelligence, but also educational outcomes, school safety, and promote social connection. However, there is no standard for when and how SEL models are applied. This paper offers suggestions on how to standardize SEL models through three aspects: educational, developmental, and social. Standardizing SEL models in this way is likely to improve learning outcomes and provide Americans with a skillset to create quality relationships across their lifespans. Building quality relationships in this way should reduce or even eliminate the trend toward loneliness and isolation.

Keywords: Emotional intelligence, Social Emotional Learning, Loneliness, Isolation

Our Epidemic of Loneliness

On June 4th, 2023, the Office of the Surgeon General published a report declaring loneliness and isolation are a "profound threat to our health and well-being" (p. 5). Titled, *Our Epidemic of Loneliness and Isolation*, the report found that Americans were spending more time alone (285 minutes per day in 2003, compared to 333 minutes per day in 2020) and less time with friends in-person (60 minutes per day in 2003, compared to 20 per day in 2020) than in previous years (p. 13). If the trend toward loneliness and isolation continues, the Surgeon General estimated the risk of both heart disease and stroke will increase by 32%. Risk of premature death will increase by 26 to 29%. Additionally, the Surgeon General estimated economic and education loss resulting from loneliness. A lack of social connection among the elderly will increase Medicare spending by $6.7 billion (p. 9). Workplace absences attributed to a lack of social connection are expected to result in economic losses of $ 154 billion (p. 9). As for schools, a lack of social connection was found to be associated with reduced academic achievement (p. 9). Taken together, the deleterious effect of loneliness will span physical to mental health, academic to vocational performance, and institutions for physical health to economic health.

According to the Surgeon General, social connection remediates loneliness. Social connection was defined as the product of three factors: structure, function, and quality. Structure included "The number and variety of relationships and frequency of interactions" (p. 11). Function was "The degree to which relationships serve various needs" (p. 11). Finally, quality was measured as "The positive and negative aspects of relationships and interaction" (p. 11). The most valuable social connections come from close personal relationships. Labeled as "key data," the Surgeon General's report noted that people find the most meaning, purpose, and

motivation in “social relationships, particularly with family and close friends” (p. 33). Quality relationships with family and close friends were estimated to have a positive effect on biology (e.g., stress hormones, inflammation), psychology (e.g., meaning, resilience), behavior (e.g., exercise, nutrition), and were inversely related to factors of morbidity (e.g., heart disease, stroke, premature death) (p. 31). Given the Surgeon General’s findings, it is imperative that Americans are provided with the necessary skills to develop social connections across the lifespan. One way this can be accomplished is to improve emotional intelligence by implementing SEL models in early childhood education.

Emotional Intelligence

Researchers Peter Salovey and John Mayer (1990) defined emotional intelligence as an individual’s capacity to track feelings and emotions in self and others, and to use this information to modify thinking and actions in the self. Emotional intelligence has since been found to promote life satisfaction and self-esteem in adults (Carmeli et al., 2009), help college students identify and respond to negative emotions (Barrett et al., 2001), and was found to predict marriage satisfaction and conflict resolution in heterosexual couples (Smith et al., 2008). Thus, emotional intelligence may be leveraged to improve relationship satisfaction, the result of which should be increased social connection.

Importantly, the benefits of emotional intelligence are not limited to adults. Researchers in Italy found that emotional recognition, a facet of emotional intelligence, was predictive of fluid intelligence in children between the ages of 8 and 12 (Agnoli et al., 2019). Fluid intelligence is important for creative problem solving, especially in mathematics (L. Ellis et al., 2018). Emotional intelligence is also positively associated with school connectedness (Iqbal et al., 2021) which, for each point increase, has been found to reduce occurrences of bullying by

4% (Carney et al., 2022). School connectedness is the belief by students that their peers and teachers care about both their academic and personal wellbeing (Centers for Disease Control and Prevention, 2009). These two points are perhaps the most important because relationship problems are common among individuals who commit violent acts at school. Key findings from a 2019 report on school violence by the U.S. Secret Service determined that individuals who committed violent acts at school experience relationship issues with peers (63%), usually from bullying (46%), and romantic partners (22%) (Alathari et al., 2019). By improving emotional intelligence, students are likely to develop better relationships with their peers and their teachers. Improved social connections in this way may even improve academic outcomes and be protective of student safety. Better relationships and safer learning environments are likely to increase social connection, thereby reducing feelings of isolation and loneliness.

One last word on the benefits of emotional intelligence. While identifying emotions can explain school connectedness generally, this effect has been found to be stronger in boys than in girls (Kurniawan & Dewi, 2016). Because externalizing behavior is more prominent in boys (Nivette et al., 2019), a stronger effect of school connectedness among males should be even more beneficial to facilitating a safe learning environment. Finally, evidence suggests recognizing emotions is predictive of behavioral adjustment cross-culturally (Barrett et al., 2001; Yoo et al., 2006). Thus, despite emotional intelligence differentially benefitting males, in the context of school violence, all students will benefit from an intervention that improves emotional intelligence. That emotional intelligence is beneficial cross-culturally strengthens its potential as an intervention to promote social connection in diverse settings.

Given the evidence that emotional intelligence can be leveraged to improve social connection, the balance of this paper will be dedicated to proposing a standardized method for

applying SEL models in the U.S. education system. To justify this approach, a review of the SEL literature will be conducted, followed by examination of a specific SEL model known as RULER. The paper will conclude with recommendations on incorporating elements such as vocabulary learning during critical childhood development periods, and structured discussion which can be leveraged to target emotional intelligence.

SEL Models and Social Connection

Studies indicate that SEL can accomplish the goals laid out in this paper. Researchers Mahoney, Durlak, and Weissberg (2018) conducted a meta-analysis of more than 200 journal articles on SEL published from 1955 to 2014. Their analysis revealed two major findings:

> Compared to control students, students participating in SEL programs showed significantly more positive outcomes with respect to enhanced SEL skills, attitudes, positive social behavior, and academic performance, and significantly lower levels of conduct problems and emotional distress.
>
> The higher academic performance of SEL program participants translated into an 11 percentile-point gain in achievement, suggesting that SEL programs tend to bolster, rather than detract from, students' academic success (p. 20).

By contrasting control with experimental conditions, the authors provide compelling evidence that social learning is an improvement over traditional learning models, and across multiple domains. These findings were further corroborated by researchers Durlak, Mahoney, and Boyle (2022) who examined 12 meta-analyses involving 523 unique studies of an estimated one-million students. In both the Mahoney et al. (2018) and Durlak et al. (2022), the effect sizes ranged from very small to large, indicating some SEL interventions are better than others. Regardless of the model, the strongest effects

were seen in SEL skills, conduct, emotional competency, and academic performance. Taken together, these studies demonstrate that SEL models offer meaningful improvements to traditional education programs. Given the evidence reviewed thus far, improvements in these categories will increase emotional intelligence (i.e., through improved SEL skills and emotional competency) in participants and likely lead to quality relationship formation. Building quality relationships is key to reducing feelings of loneliness and isolation (Office of the Surgeon General, 2023).

Putting SEL into practice, Yale's Center for Emotional Intelligence developed a model they dubbed RULER. RULER is an acronym for Recognizing, Understanding, Labeling, Expressing, and Regulating emotions (*What Is RULER?*, n.d.). Yale researchers describe those elements as "the five skills of emotional intelligence." According to the Sample RULER Elementary Teaching Guide (2021) available online, these skills are to be applied multimodally (e.g. creative writing, art work, class discussion), and the curriculum includes class and at home exercises. Assignments require students use age-appropriate emotional vocabulary (e.g., for Kindergartners: annoyed, hurt, relaxed, excited) (p. 13). Class activities require students to employ these words in reflection assignments and in group discussions, with occasional check-ins with the teacher for understanding, and a complement of breathing exercises to regulate emotions and build compassion in self and others (p. 14). In total, the elementary school program includes 18 units each focusing on one element of the RULER acronym and stages of emotional vocabulary learning and application (p. 15-16). According to the RULER web page, this program improves school climate (i.e. more emotional support, better instruction), increases academic performance (i.e. RULER participants had higher grades compared to control groups), increases relationship quality between peers and between students and teachers, and reduces

occurrences of bullying and aggressive behavior compared to control groups (*Evidence for RULER*, n.d.).

To contextualize the claims from the RULER website, three studies have been selected from the web page for brief review. The first study by researchers Brackett et al. (2012) found students who participated in RULER had higher year end grades, rated their teachers more favorably, and demonstrated higher emotional competence when compared to students in a control group. The second, by Cipriano et al. (2019), discovered at risk students who participated in a RULER program were more engaged in class, and demonstrated improved classroom conduct a year later when compared to similar students who did not participate. Finally, Hagelskamp et al. (2013) found, compared to traditional classrooms, RULER classrooms exhibited increased emotional support, better classroom organization, and increased instructional support. These improvements persisted a year later and were found to be partially explained by participation in the RULER program. Thus, by improving emotional intelligence, the RULER program has provided further evidence that SEL models are an improvement to existing educational models.

Of course, when considering a new program, one must consider costs as well as benefits. Pricing for RULER training varies by number of participants and training medium. For online training, 3 to 5 participants ranges in cost from $6,000 to $8,000 (*RULER Institute Online*, n.d.). For in person training, 3 to 5 participants ranges in cost from $7,500 to $10,500 (*RULER Institute at Yale*, n.d.). Given the projected benefits across the lifespan, and the positive effect on education outcomes, relationships, and mental health, the benefits are likely to outweigh the costs. For low socio-economic schools that may find it difficult to pay for this training, perhaps a federal initiative can be advanced to fund such training.

Lastly, there are some possible concerns with the RULER model. In the Cipriano et al. (2019) study, RULER was negatively associated with performance of black students. Given that white and Hispanic students benefitted at the cost of black students, this program should be further tested before implementing on a large scale. Though the researchers did not offer an explanation for this difference, they noted the data for this study was collected from 2007 to 2010, and pointed out the program had gone through significant changes in the intervening years. Nevertheless, future studies must assess SEL models for differential outcomes with respect to race. By identifying the reason for these differences, researchers can address and correct them before implementing SEL models broadly.

Standardizing SEL Frameworks

Researchers Mahoney, Durlak, and Weissberg (2018) concluded the problem of SEL models is not what they deliver (e.g., academic performance, improved social connection) but ensuring the correct model is in place. Proposed here is a standardized framework for structuring SEL models. This framework considers three elements: educational, developmental, and social. The educational aspect includes social learning informed by Zone of Proximal Distance (ZPD) and teaching a baseline of emotional vocabulary. The developmental aspect suggests the most effective time to employ a vocabulary-based model is in early childhood. Finally, the social aspect structures conversations by pairing Rational Emotional Behavioral Therapy (REBT) and symbolic play with inductive discipline.

The general idea is as follows. Standardizing SEL models with these elements will create a universal emotional intelligence toolset. In terms of education, ZPD harnesses the human element of education to reinforce learning, and providing children with a basic emotional vocabulary will create a sense of shared emotional literacy, enhancing understanding when

interlocutors share emotional states. Employing this education element during a critical period for language development should ensure the vocabulary is assimilated early, quickly, and available for use across the life span. Nesting both the educational and developmental aspects within therapeutic interventions, play, and discipline should inculcate at the individual level a method by which to evaluate and regulate emotions in a healthy way. The educational, developmental, and social aspects are discussed under their respective headings.

Educational

SEL and Zone of Proximal Distance

When Peter Salovey and John Mayer (1990) coined the term *emotional intelligence*, they described it as "a set of skills" (p. 185). Though emotional intelligence is a skill set, it has roots in biology. A review of the literature on emotional intelligence suggests females more accurately identify emotions, while males tend to rate emotions with greater intensity, especially anger, which invokes a fight-or-flight response more often (Kret & De Gelder, 2012). Researchers have found similar results in adolescents (D'Amico & Geraci, 2022). However, emotional intelligence has also been found to increase with age (Van Rooy et al., 2005). Taken together, these findings suggest SEL models should emphasize helping males more accurately interpret emotions by reducing perception of threat when threat is not present. Ideally, SEL models that accomplish that goal will further enhance the skills of students who already rate high in emotional intelligence. Because emotional intelligence increases with age, to accomplish these goals, SEL models should leverage adult mediated social interactions.

Such interactions in education are known as ZPD. ZPD was proposed by psychologist Lev Vygotsky (1978). The distance to which Vygotsky referred was between what a student knows about a problem and what they need to know to solve it. The role of the teacher is to

provide the minimal guidance necessary to bridge that distance. Informed by ZPD, educators have developed an active learning strategy dubbed *Think, Pair, Share*, the goal of which is to have students discuss a topic and report back to the class, while the teacher poses questions to facilitate learning (Lightner & Tomaswick, 2017). SEL models should make similar use of ZPD and *Think, Pair, Share*.

Evidence suggests active learning strategies like *Think, Pair, Share* can be used to improve vocabulary. In a study involving 27 Head Start classrooms, researchers Hindman, Wasik, and Bradley (2019) recorded teacher-student interactions during group reading sessions. After analyzing the videos, the researchers discovered that when teachers paused to ask students questions, it strengthened vocabulary skills in high vocabulary skill students, and improved learning in low vocabulary skill students. If this is the case, then pairing emotional vocabulary with storytelling and questioning should result in improved emotional vocabulary. Emotional vocabulary and social communication skills have been found to be associated with reduced behavioral problems (Rimehaug & Kårstad, 2022). Through ZPD and active learning strategies during reading sessions that focus on emotion, students will learn emotional vocabulary and benefit from social communication to reinforce their learning. In theory, this will improve the learning environment by reducing behavioral problems, and an improved learning environment will likely facilitate social connections, further reducing feelings of loneliness and isolation.

There is evidence to suggest this is true. Researchers studying emotional regulation in children between the ages of 3 and 6 found that verbal skill, non-verbal intelligence, and emotional comprehension were related to emotion regulation strategies, and that these effects improved with age (Sala et al., 2014). Together, these studies suggest that social interactions improved vocabulary learning, and that pairing emotional vocabulary with social communication

could be used to reduce behavioral problems in the classroom. As children grow older, the effect on emotional regulation is likely to improve.

Baseline Vocabulary

In terms of vocabulary, SEL models should focus on teaching words used to measure emotion from the Discrete Emotions Questionnaire (DEQ). Developed by Harmon-Jones et al. (2016), the DEQ maps basic emotional vocabulary to eight categories of emotions, which are differentially associated with approach (e.g., anger motivates action, whereas sadness depresses action). These categories include anger, anxiety, desire, disgust, fear, happiness, relaxation, and sadness. A series of experiments were conducted in which participants were asked to evaluate emotionally salient stories and provide written responses. From the written responses, researchers derived a list of the most frequently used words and selected the most accessible words from that list. The resulting list formed the DEQ, and has since been used in a host of studies including measuring emotional states resulting from playing video games (Bonk & Kim, 2022), behavioral neuroscience involving serotonin and emotion regulation (Bîlc et al., 2023), and to assess the relationship between civic engagement and mental health in adults (Fenn et al., 2024).

Having demonstrated validity across domains, and given the accessibility of the vocabulary, and the mapping of the vocabulary to approach motivation, the vocabulary from the DEQ will be a useful baseline for emotional vocabulary. However, some of the more intense descriptions (e.g., ‘pissed off’) may need to be adjusted for a younger audience. The entire list of words can be found in the supplemental materials section of the Harmon-Jones et al. (2016) article. As students progress through the educational system, their vocabulary should be expanded to account for more complex and even nuanced emotional language. To address this

need, authors Adler, Proctor II, and Manning (2023) devote an entire chapter to emotions in their communications textbook, *Looking Out, Looking In* (see chapter 5, table 5.1). Combined, a base emotional vocabulary in early childhood will provide an analog for emotional conveyance broadly, and an expanded vocabulary, if pursued, can add nuance and maintain interest by adding novel words.

Developmental

Modified Vocabulary Learning

A key element of emotional intelligence is identifying emotional states. Research suggests that a child's vocabulary affects their ability to identify emotional states. Grosse and Streubel (2024) developed a model to measure emotional vocabulary and emotional recognition. They found that general vocabulary predicted emotion recognition in 10- to 11-year-olds, whereas emotional vocabulary better predicted emotion recognition in 16- to 17-year-olds. Though the study had a relatively small sample (57), their finding that vocabulary and emotion recognition has been replicated in many other studies (Beck et al., 2012; Griffiths et al., 2020; Pons et al., 2003). Therefore, improving both general and emotional vocabulary is one way to improve emotional intelligence.

The best time to emphasize vocabulary learning is in early childhood education. Children learn language at an exponential rate, with an estimated vocabulary of 60,000 words by the time they are adults (McMurray, 2007). Children learn about 10,000 of those words by age 5, and up to half by age 6 (Berger, 2021; Shipley & McAfee, 2023). Given that kindergarten begins at age 5, early education provides an ideal opportunity to introduce emotional vocabulary. Therefore, SEL models emphasizing emotional language learning will be most effective if implemented in early education.

Social

Rational Emotive Behavioral Therapy & Symbolic Play

When students engage in class discussion, those discussions should include three elements: Rational Emotive Behavioral Therapy (REBT), inductive discipline, and symbolic play. REBT was developed by Albert Ellis (1958) as a form of cognitive behavioral therapy where individuals are taught to rationally examine their feelings. Ellis described the role of the therapist as a counter to negative thinking, pointing out contradictions to negative thoughts, and to encourage patients to participate in an activity to reverse their negative thoughts. Over the last 50 years, REBT has been found to be a valid psychological intervention in treating irrational beliefs (David et al., 2018). In some cases, it has been used to improve social skills in children diagnosed with autism-spectrum disorder (Nnamani et al., 2019; You et al., 2024). To be clear, the idea here is not to clinically diagnose children with autism spectrum disorder or to convert teachers into therapists so that they may treat children for irrational beliefs. Instead, the idea is to use treatments that successfully demonstrate emotional regulation in clinical settings in an academic setting.

The goal is to create stories that use REBT methods to teach children to analyze their emotions through symbolic play. An overview of symbolic play and the stages in which it manifests during childhood can be found on the California Department of Education's (2024) Infant / Toddler Development website. The agency defines symbolic play as a child's "ability to use actions, objects, or ideas to represent other actions, objects, or ideas." At about 36 months, children can imagine themselves as characters in dramatic play. Thus, SEL models should have children make inferences about their behavior through storybook characters engaged in dramatic play. For example, one character may observe another is angry. The angry character will be

dismissive of the observer. The observer will then be portrayed as upset, and the events take a minor spiral. A third character will intervene and engage in the process of clarifying why the first character was upset and ask the observing character to examine their original interpretation by taking another perspective. The interaction will make use of emotional vocabulary that pairs with the emotions expressed. Children will be asked to verbally role play these scenes to get to the same conclusion as in the story.

This is where symbolic play offers significant value. Symbolic play and language development have been considered highly related by multiple studies (DeLoache, 2002; McCune, 2009; Orr & Geva, 2015; Smith & Jones, 2011). From REBT, the characters will engage in activities that match Ellis's ABC model: (A) activating event, (B) irrational belief, (C) consequences of irrational belief (Dobson, 2010, p. 12). Effectively, children will play the role of the therapist and the patient in these stories. Ideally, children will be engaged socially, which will reinforce vocabulary learning, and see it as a form of play, further enhancing language skills, and do so within a clinically proven framework of REBT. Because REBT has been found effective at improving the social skills of children with autism spectrum disorder, this model should be effective for individuals who rate high or low on social skills. More importantly, children will be provided with not only emotional vocabulary, but a clinically proven toolset to interpret negative emotions.

Inductive Discipline

The overarching theme of this framework is to reduce externalizing behavior in children to provide a safe learning environment and facilitate relationship formation. To do this effectively, teachers must be equipped with disciplinary training. This is where inductive discipline comes in. Inductive discipline is, effectively, the practice of bargaining with children.

It relies on setting limits, reminding children of rules, and reasoning with children about their conduct and the consequences of their actions (Dodge et al., 1994; Hart et al., 1992; Hoffman, 2000). In a longitudinal study, researchers Choe, Olson, and Sameroff (2013) studied the relationship between inductive discipline, physical discipline, and behavioral problems in children. The sample included 241 children, 49% of whom were girls, and 86% were European American. The remainder were bi-racial. At ages 3, 5.5, and 10, the mothers and teachers of the children completed various surveys measuring inductive discipline, physical discipline, and externalizing problems. The results indicate physical discipline increased externalizing problems across the age range, while inductive discipline reduced externalizing problems.

The goal here is to use modeling paired with inductive discipline in stories, as well as a strategy for teaching. Combining these elements in stories might further reduce behavioral issues in the classroom and improve the learning environment. Taking a cue from the Choe et al. (2013) paper, the relationship dynamic between the behavioral models and the main characters should be that of parent and child. Students will be tasked with reflecting on instances of inductive discipline. The idea here is that students will be absorbing a disciplinary structure (inductive discipline) and an emotional processing treatment (REBT), which incorporates emotional vocabulary that has been used to measure both emotion and motivation. Together, these practices from across psychology should provide students with a robust toolset to interpret and regulate emotions.

REBT & Discussion

Nesting conversation about emotions within symbolic play and REBT will enhance SEL effectiveness in early education. In the 3rd edition of the *Handbook of Cognitive Behavioral Therapies* (Dobson, 2010), clients are meant to describe what event activated the emotion, what

they believe happened, their emotional behavior, and the consequences that follow (p. 245). Students in the classroom should be asked to reflect on these same elements using symbolic play as a form of perspective taking. Perspective taking is defined by the American Psychological Association (2018) as "looking at a situation from a viewpoint different from one's usual viewpoint" which "may involve adopting the perspective of another person … in role play exercises". This would include asking children questions like "If you were this character, how might this make you feel?" or "Given what happened, do you think this character was right to feel this way?". Applied perspective taking similar to this has been used effectively to improve social skills in individuals diagnosed with autism spectrum disorder (Peters & Thompson, 2018), and will likely generalize to neurotypical participants. The combination of REBT emotional assessment paired with perspective taking as a form of symbolic play should help children understand the role of emotions in their lives by adding elements of self-reference, regardless of their level of social skills.

Case for a Standardized SEL Model

Standardizing SEL models will benefit social connection. By applying SEL models at a developmentally advantageous life stage (i.e., early childhood education), students will have access to a common emotional vocabulary capable of conveying motivated states, which can be interpreted by educators and therapists alike. Infusing social interactions with therapeutic elements, symbolic play, and discipline will provide children with a structured social situation in which emotional states can be expressed as a game, which simultaneously reduces the stakes of the interaction but also promotes learning. Inductive discipline can be further used by educators to help children regulate emotional expression. Each of these elements develops different facets of emotional intelligence from emotional recognition to expression to regulation. Standardizing

SEL models in this way would provide a minimum skillset that improves social connection, education outcomes, and safer learning environments broadly. Furthermore, these modifications would still allow for unique SEL models, like RULER. Existing models would still be able to tailor modules or assignments according to their pedagogical inclinations, allowing for a diverse range of models. Standardizing developmental, educational, and social aspects would simply ensure consistency.

Conclusion

By improving emotional intelligence in early education, students will learn skills they need to make quality relationships and end the epidemic of loneliness. If left unaddressed, a lack of social connection will have a deleterious effect on the physical and mental health of Americans. The economy could suffer billions of dollars in lost productivity. Medicare will be encumbered by additional costs. Educational outcomes will decline. But all is not lost. Quality relationships combat the effects of loneliness. The evidence suggests that emotional intelligence is instrumental in building quality relationships. In adulthood, emotional intelligence was linked to improved conflict resolution and better quality interpersonal and romantic relationships. Further evidence suggested that SEL programs throughout childhood and adolescence provided children with the skills they needed to improve emotional intelligence. In theory, honing these skills in childhood will improve relationship quality throughout the lifespan, reduce school violence, improve grades, and lead to better socially and emotionally adjusted children. By implementing standardized SEL programs, Americans will benefit from improved emotional intelligence at a developmentally advantageous stage, will be better educated, better adjusted socially, and happier people. Though emotional intelligence may not be a treatment for all that

ails Americans, the evidence suggests it will be an effective treatment for one of the most pressing ailments: The epidemic of loneliness.

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